

# CHORD: Composable Hybrid Optical Reconfigurable Diffractive Framework For Optical Neural Network

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**Abstract**— Diffractive optical neural networks (DONNs), leveraging free-space light wave propagation for ultra-parallel, high-efficiency computing, have emerged as promising artificial intelligence (AI) accelerators. However, their inherent lack of reconfigurability due to fixed optical structures post-fabrication hinders practical deployment in the face of dynamic AI workloads and evolving applications. To overcome this challenge, we introduce, *for the first time*, a composable hybrid optical reconfigurable diffractive framework (CHORD), a physically composable architecture that unlocks a new degree of freedom and unprecedented versatility in DONNs. By leveraging full-system learnability, CHORD repurposes fixed fabricated optical hardware, achieving exponentially expanded functionality and superior task adaptability through the differentiable learning of system variables. Furthermore, CHORD adopts a hybrid optical/photonic design, combining the reconfigurability of integrated photonics with the ultra-parallelism of free-space diffractive systems. Extensive evaluations demonstrate that CHORD has digital-comparable accuracy on various task adaptations with  $74\times$  faster speed and  $194\times$  lower energy. Compared to prior DONNs, CHORD shows exponentially larger functional space with  $5\times$  faster training speed, paving the way for a new paradigm of versatile, composable, hybrid optical/photonic AI computing. Our code is open-sourced at [link](https://github.com/ScopeX-ASU/CHORD)<sup>1</sup>.

## I. INTRODUCTION

Emerging technologies promise efficiency and performance breakthroughs, and optical neural networks (ONNs) are at the forefront of this movement. They are poised to revolutionize next-generation AI computing by harnessing the inherent parallelism and speed of light. Diffractive ONNs (DONNs) [1], [2], [3], [4], [5] represent a compelling realization of optical AI hardware, employing free-space optical signal modulation, diffraction, and interference to achieve ultra-parallel intelligent information processing. Different from integrated photonic tensor cores (PTCs) [6], [7], [8], [9], [10], [11], [12] based on programmable photonic integrated circuits (PICs) designed for matrix-vector multiplication, free-space DONNs eliminate the dimension limitations of 2-D silicon chips and operate directly on light waves in 3-D free space, offering unprecedented parallelism through a global-view spatial linear operation.

However, DONNs face a significant challenge in their lack of reconfigurability after fabrication. The fixed manufactured phase masks limit their adaptability to new ML tasks, hindering their deployment in dynamic, real-world AI workloads [13], [14]. While research into reconfigurable phase masks based on tunable metasurfaces shows promise, these approaches remain in the immature stages. They struggle with limitations such as low endurance and reliability, which make them unsuitable for adoption in the near term.

To address these limitations, we shift our focus from device-level reconfigurability to system-level reconfigurability. Optical systems inherently feature adjustable parameters, such as wavelength, phase mask spacing, and element size, that significantly influence system behavior but are often manually determined based on experience in the early stage and fixed during the optimization process later. This raises a key question: **can these built-in hardware variables be automatically optimized for better expressivity?** However, for true task adaptation, adjusting these scalar variables alone lacks sufficient degrees of freedom. A fundamental system redesign is necessary to

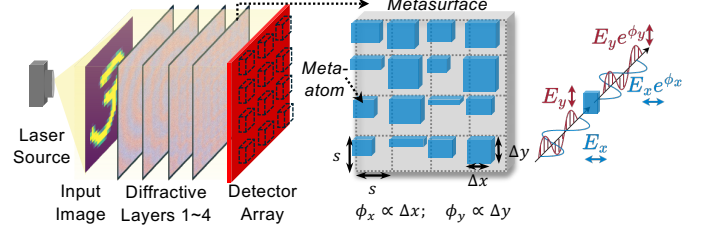


Fig. 1: Diffractive optical neural networks (DONN) based on metasurfaces with a pixel size of  $s$ . The rectangular meta-atom shape can be designed to realize a polarization-dependent phase shift for orthogonally polarized light.

unlock the versatility needed for modern AI workloads. In this work, we address this challenge: **how to construct a versatile DONN system with exponentially many reconfigurable functionalities by re-purposing the same set of fabricated phase masks?** Furthermore, considering the strengths of both diffractive ONNs and integrated PTCs, we explore **how to hybridize free-space optics and integrated photonics organically** to leverage both the ultra-parallel processing capability of DONNs and the superior reconfigurability of integrated PTCs for joint NN acceleration.

To answer those questions, we investigate the intrinsic learnability of DONNs from the system level and propose a multi-dimensional reconfigurable hybrid DONN design paradigm, dubbed CHORD, with a physically composable architecture that can flexibly switch the orientation and placement order of phase masks for exponentially larger functional space. The main contributions are as follows,

- We present *the first* in-depth analysis of multi-dimensional learnability in DONNs, introducing a physically composable hybrid optical system CHORD for extreme multi-functionality.
- **Auto-Learned Multi-Dimensional System Variables:** we enable differentiable learning of DONNs across multiple dimensions (wavelength, spacing, orientation, permutation), exponentially enhancing expressivity and the functional space.
- **Hybrid Optical/Photonic System:** We explore the possibility of hybridizing free-space diffractive optics and reconfigurable integrated photonics for joint NN acceleration.
- **Superior Task Adaptability:** Extensive evaluations show that our system and differentiable learning algorithms enable digital-comparable task adaptation performance with  $74\times$  faster speed and  $194\times$  lower energy, outperforming prior DONNs with exponentially higher functional space and  $5\times$  higher training efficiency.

## II. BACKGROUND

### A. Diffractive Optical Neural Networks (DONN)

As shown in Fig. 1, the main trainable parameters are weights on the phase mask, i.e.,  $\Phi \in \mathbb{R}^{K \times K}$ , where the phase shift of one unit cell is  $\phi_{i,j} \in [0, 2\pi)$ . When incident light  $X$  passes through the phase mask, it applies pixel-wise phase rotation, i.e.,  $e^{\Phi} \odot X$ , where  $\odot$  is element-

<sup>1</sup><https://github.com/ScopeX-ASU/CHORD>

wise multiplication. Typically, to increase the number of trainable parameters, a DONN will have multiple cascaded phase masks, e.g.,  $L$  layers [15]. Between two masks, the light will propagate in the free space with diffraction. This can be described using the Rayleigh-Sommerfeld diffraction, a dense linear transformation where each light on the output plane is the superposition of all light from the input plane with a distance-related intensity/phase modulation coefficient  $h$ . The transmission, including a phase mask and diffraction, is denoted as  $X_{l+1} = \mathcal{H}(e^{\Phi_l} \odot X_l)$ , formulated as [16]:

$$X_{l+1}(k_x^{l+1}, k_y^{l+1}) = \sum_{k_x} \sum_{k_y} h_l(R, z, \lambda) \cdot e^{\phi_{k_x^l, k_y^l}} \cdot X_l(k_x^l, k_y^l), \quad (1)$$

$$h(R, z, \lambda) = \frac{1}{2\pi} \frac{z}{R} \left( \frac{1}{R} - j\mathbf{k} \right) \frac{e^{jkR}}{R},$$

$$R = \sqrt{((k_x^{l+1} - k_x^l) \cdot s)^2 + ((k_y^{l+1} - k_y^l) \cdot s)^2 + z^2}.$$

In the diffraction coefficient  $h$ ,  $\mathbf{k} = \frac{2\pi}{\lambda}$  is the wave vector in free space, and  $\lambda$  is the wavelength, and  $z$  is the distance between two layers,  $j = \sqrt{-1}$ .  $R$  is the distance from the unit cell located at  $(k_x^l, k_y^l)$  in the  $l$ -th layer to the unit cell located at  $(k_x^{l+1}, k_y^{l+1})$  in the  $(l+1)$ -th layer.  $s$  is the pixel size of the phase mask unit cell.

As nanofabrication technology advances, a promising approach to implement DONN is using a sub-wavelength metasurface. Metasurfaces offer a compact solution, where each unit cell is called a meta-atom and consists of sub-wavelength nanostructures [2], [17], [18]. A typical meta-atom is a rectangular nano-pillar with precisely defined width, length, and thickness to achieve a specific phase shift, in Fig. 1. This structure is polarization-dependent and thus can induce independent phase changes in orthogonal light polarizations, e.g.,  $\phi_x$  for x-polarized and  $\phi_y$  for y-polarized light. This is the foundation of our polarization-differential computing technique introduced later.

### B. Integrated Photonic Tensor Cores

A different approach to implementing ONNs is through photonic tensor cores (PTCs). Unlike DONNs based on passive diffractive components, PTCs integrate photonic devices to actively modulate light to achieve high-speed matrix-vector multiplication (MVM) [6], [19], [20], [21], [11], [9], [10], [22]. DONNs are more natural for ultra-parallel low-energy spatial processing, while PTCs are more suitable in generic reconfigurable MVMs. This indicates complementary capabilities between the two hardware platforms for joint computing.

## III. RELATED WORKS

Previous works have proposed solutions to build multi-functional DONNs. For instance, prior methods have introduced rotation into diffractive neural networks, enabling certain layers to be rotatable and thereby enhancing task adaptability [13], [14]. Prior work [18] also explored training different tasks on separate regions of the diffractive layer. This approach suffers from area limitations, constrained to a limited number of fixed functions (e.g., 2-3). Similarly, metasurface-based DONNs [17] leverage polarization-dependent phase responses to train multiple tasks, with each task corresponding to a different polarization state that the metasurface can distinguish. This approach faces limited expressivity as only a small number of orthogonal polarization states are supported by metasurfaces.

## IV. PROPOSED COMPOSABLE HYBRID OPTICAL RECONFIGURABLE DIFFRACTIVE FRAMEWORK CHORD

We introduce a hybrid DONN with multi-dimensional reconfigurability and physical composability, dubbed CHORD (see Fig. 2). By delving

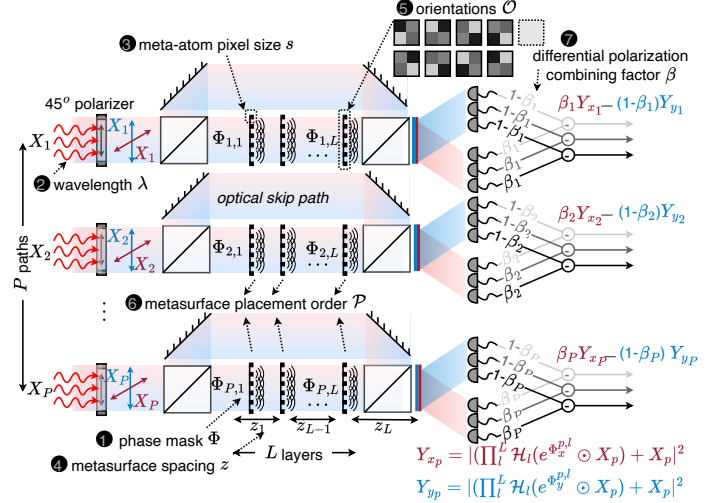


Fig. 2: Multi-path diffractive layer (DiffLayer) with  $P$  paths. Each has  $L$  cascaded metasurfaces and polarization-differential photodetection. Learnable parameters 1-7 are marked.

into the system-level learnability, CHORD introduces new degrees of freedom to overcome the reconfigurability limitations of traditional DONNs, enabling superior task adaptability and training efficiency.

TABLE I: Parameter tuning mechanism (structural, electrical, mechanical) and reconfiguration frequency. 1-7 belong to diffractive optical hardware. 8 is on integrated photonic chips.

| Learnable System Parameters                            | Stru. | Elec. | Mech. | Reconfig. Freq.  |
|--------------------------------------------------------|-------|-------|-------|------------------|
| 1 Phase Response $\Phi$                                | ✓     |       |       | Fixed after fab. |
| 2 Source Laser Wavelength $\lambda$                    |       | ✓     |       | Tunable per task |
| 3 Meta-atom Pixel Size $s$                             | ✓     |       |       | Fixed after fab. |
| 4 Metasurface Spacing $z$                              |       |       | ✓     | Tunable per task |
| 5 Orientation $\mathcal{O}$                            |       |       | ✓     | Tunable per task |
| 6 Placement Order $\mathcal{P}$                        |       |       | ✓     | Tunable per task |
| 7 Differential Polarization Factor $\beta$             |       | ✓     |       | Runtime tunable  |
| 8 Channel-Mixing Factors $\alpha_{pre}, \alpha_{post}$ |       | ✓     |       | Runtime tunable  |

### A. Overview of Learnable System Parameters

For a generic expressive DONN, our CHORD is assumed to have a **multi-path** architecture with  $P$  parallel paths, each with  $L$  cascaded phase masks, shown in Fig. 2. We summarize programming mechanisms and properties of all variables in our CHORD architecture in Table I, including the *conventional* parameter phase masks  $\Phi$ , and our **7 newly introduced dimensionalities**: light source wavelength  $\lambda$ , meta-atom pixel size  $s$ , diffraction distance between metasurfaces  $z$ , metasurface orientation  $\mathcal{O}$ , metasurface placement ordering  $\mathcal{P}$ , pre- and post-channel-mixing factor  $\alpha_{pre}$  and  $\alpha_{post}$ , and differential polarization combining factor  $\beta$ , visualized in Fig. 2.

1) **1 Phase Masks  $\Phi$** : Phase masks are the core parameters of DONNs. If phase elements (meta-atoms) are designed to be polarization-dependent, e.g., a rectangle, we can adjust meta-atom width/length to independently change its phase responses  $\phi_x, \phi_y \in [0, 2\pi)$  for x- and y-polarized light. Once a phase mask is trained and fabricated, it is fixed and typically cannot be reconfigured.

2) **2 Light Source Wavelength  $\lambda$** : The wavelength  $\lambda$  impacts the optical system's diffraction behavior  $\mathcal{H}(\cdot|\lambda)$  while maintaining a near-constant phase response  $\phi$  if using broadband meta-atoms. A reasonable range for  $\lambda$  is from 400 nm in the visible spectrum to 1600 nm in the near-infrared. Given a preferred  $\lambda$ , e.g., 532 nm, we consider local fine-tuning with  $\pm 20$  nm linewidth given the tunability of the current laser with a 0.1 nm tuning resolution.

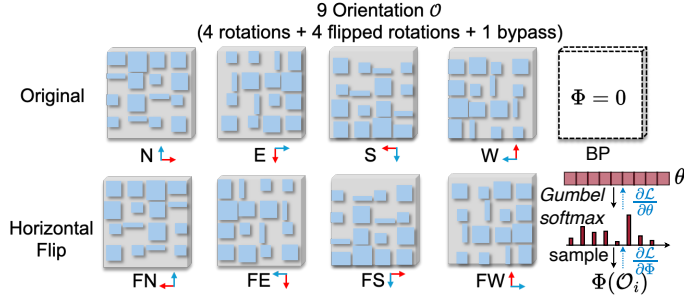


Fig. 3: 9 orientation states for each bi-directional phase mask, which can be learned differentially using Gumbel-softmaxd.

3) ④ **Meta-Atom Pixel Size  $s$** : The diffraction operator  $\mathcal{H}(\cdot)$  is a function of pixel size  $s$ , which is the period of the metasurface grid. Considering all metasurfaces have a shared scalar pixel size  $s \in \mathbb{R}$ , e.g., 300 nm, we are allowed to freely determine this variable during the initial design stage for better performance. The pixel size is lower-bounded by the meta-atom width required to realize  $2\pi$  phase response plus minimum manufacturable linewidth, e.g.,  $s > s_{2\pi, \lambda_0} \cdot \lambda / \lambda_0 + \Delta s$ . For example, given a reference  $\lambda_0 = 532$  nm, the meta-atom width needs to be  $s_{2\pi, \lambda_0} = 180$  nm to realize  $2\pi$  phase shift, then we assume a linear scaling w.r.t.  $\lambda$ . With a  $\Delta s = 20$  nm gap between pixels, the pixel period needs to honor the constraint  $s > 200$  nm.

4) ④ **Metasurface Spacing  $z$** : The spacing between two phase masks determines the diffraction behavior of the light, i.e.,  $\mathcal{H}(\cdot|z)$ . Intuitively, a small  $z$  passes the light directly to the next metasurface with weak diffraction and cross-pixel interactions, yielding a local linear operator with a nearly diagonal transfer matrix. A large spacing introduces all-to-all pixel interaction. However, most light spreads to spaces outside the next metasurface plane, which causes significant light energy loss and, thus, a dim and blurred image projected onto the next metasurface. Hence, a carefully optimized diffraction spacing  $z$  is critical to the receptive field and expressivity of the DONN. We explore a shared learnable spacing  $z$  for all layers and independently adjustable spacings ( $z_1, z_2, \dots, z_L$ ) for different layers.

5) ⑤ **Metasurface Orientation  $\mathcal{O}$** : Before metasurface fabrication, the phase mask is freely optimized to realize an arbitrary phase distribution. After fabrication, the metasurface is fixed. Thanks to the physical composability of our CHORD design, the phase mask orientation becomes an additional degree of freedom to enrich the functionality while reusing the same piece of hardware, as shown in Fig. 3. The phase mask can be rotated at four angles, i.e., R0 (N), R90 (E), R180 (S), and R270 (W). Besides, a bidirectional phase mask can also be horizontally flipped to introduce chirality, i.e., shine the light reversely through the mask. Then, we can augment to another 4 states: FN, FE, FS, and FW, shown in Fig. 3. Replacing the phase mask with an identity mask  $\Phi = 0$  is another allowable *bypass* state given the system flexibility, denoted as BP. Orientation reconfiguration can be implemented mechanically. Each metasurface can select its orientation out of 9 states. For a DONN with  $P$  parallel paths, each with  $L$  cascaded masks, there are  $9^{PL}$  unique states. Though this approach might not be suitable for real-time frequent reprogramming, it is promising for less frequent task adaptation. It enables **exponentially many new system responses for potential lifelong hardware reusing**. Later, we will introduce how to learn the best state from discrete orientations in a differentiable way.

6) ⑥ **Metasurface Placement Order  $\mathcal{P}$** : To fully leverage the physical composability of our CHORD system, we further explore the location

swapping of metasurfaces for extreme multi-functionality. Inspired by the concept of card shuffling, phase masks can be reordered and re-plugged to any valid slots. A permutation of all phase masks can be used to describe this placement ordering. In a  $P \times L$  multi-path DONN, the total permutation reaches  $(PL)!$ , enabling an even larger design space than orientation.

7) ⑦ **Differential Polarization Combining Factor  $\beta$** : Polarization-multiplexed DONN [17] is previously demonstrated to map two tasks onto orthogonal polarization channels. Differently, we use two independent polarization directions in a **differential fashion to enhance the linear operation expressivity and realize full-range outputs**. Each forward pass through our CHORD system generates two non-negative outputs  $Y_x$  and  $Y_y$  for x(y)-polarized channel, respectively. We use balanced photodetection to obtain a full-range feature map  $Y = \beta Y_x - (1 - \beta) Y_y$  with a *non-negative combining factor*  $\beta \in \mathbb{R}_+$ .  $\beta$  is electronically implemented with low overhead and thus can be dynamically reconfigured in real time.

8) ⑧ **Pre/Post Channel-Mixing Coefficients  $\alpha_{pre}$  and  $\alpha_{post}$** : Simulating such systems in a multi-channel DONN layer is time-consuming and memory-hungry. With  $C_{in}$  input feature map channels and  $C_{out}$  output channels, it requires to simulate the  $P$ -path CHORD system by  $C_{in}C_{out}/P$  times, which is not scalable. To make DONN training more scalable and efficient, we borrow the concept of depth-wise convolution, which performs **spatial-only channel-wise diffractive projection** without aggregating the computing results from  $C_{in}$  channels. To enable channel-wise feature extraction, we add a channel-mixing coefficient  $\alpha_{pre}$  and  $\alpha_{post}$  before and after the metasurface DONN layer, same as point-wise convolution. The pre-channel-mixing layer projects the features from  $C_{in}$ -channel to  $C_{mid}$ -channel. The DONN system forwards  $C_{mid}$  channels of images through the metasurface system and obtains  $C_{mid}$  output channels, which are further fed into  $\alpha_{post}$  layer and projected to  $C_{out}$  channels.

As the channel-mixing operation only involves parallel MVM, it can be efficiently mapped to integrated PTCs for ultra-fast processing with real-time reconfigurability. By leveraging the ultra-parallel global-view spatial processing capability of diffractive optical systems and ultra-fast reconfigurable MVM power of integrated photonic accelerators, our hybrid optical-photonic CHORD system can realize **highly adaptable and efficient** optical computing. Meanwhile, this depthwise separable operation, with comparable expressivity to traditional multi-channel convolution-like DONN layers, **significantly lowers the training cost** of simulating diffractive layers.

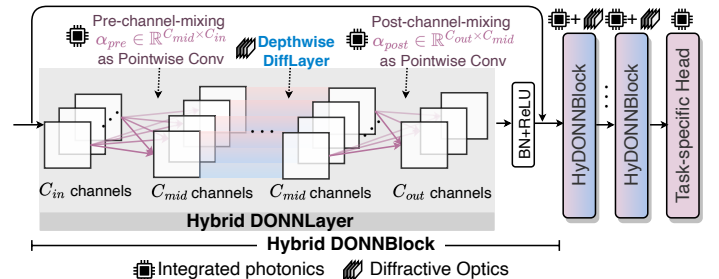


Fig. 4: Hybrid DONNLayer with pointwise Convolution mapped to integrated PTCs and depthwise diffractive layers mapped to free-space diffractive optics.

9) **Hybrid DONNLayer**: Figure 4 shows our hybrid DONNLayer with the channel-mixing operation mapped to integrated PTCs for ultra-fast point-wise Conv and diffractive layers (DiffLayer) in the middle



mapped to metasurface systems for ultra-parallel depth-wise spatial information extraction. DONNResBlock contains BatchNorm, activation, and a residual connection. Multiple DONNResBlocks construct the backbone of our CHORD system. The task-specific head at the end is mapped onto integrated PTCs.

### B. Pre-Fab Training for Initial Task

In the initial training phase before metasurface fabrication, all system variables can be freely optimized. Note that since the phase response of the metasurfaces  $\Phi$  can be freely adjusted, it is unnecessary to learn the orientation or placement order at this stage. The permutation  $\mathcal{P}$  and metasurface rotation mask  $\mathcal{M}$  will not be optimized during this phase. Therefore, we focus on optimizing parameters such as the  $\Phi$ ,  $\alpha_{pre}$ ,  $\alpha_{post}$ ,  $\beta$ ,  $\lambda$ ,  $z$ , and  $s$  on the initial machine learning task.

$$\min_{\Phi, \lambda, z, s, \alpha, \beta} \mathcal{L}((\Phi, \lambda, z, s, \alpha, \beta), \mathcal{D}_{init}). \quad (2)$$

### C. Post-Fab Training for Tasks Adaptation

Once the first training is completed, DONN phase masks will be fabricated. Phases  $\Phi$  and pixel size  $s$  are fixed. We now introduce how to enable fully-differentiable learning of the rest of the architecture variables to adapt the fabricated system to new tasks  $\mathcal{D}_{adapt}$ ,

$$\begin{aligned} \min_{\lambda, z, \alpha, \beta, \mathcal{O}, \mathcal{P}} \quad & \mathcal{L}((\lambda, z, \alpha, \beta, \mathcal{O}, \mathcal{P}), \mathcal{D}_{adapt} | \Phi^*, s^*) \\ \text{s.t.} \quad & \mathcal{P} \in \text{Permutation matrices} \\ & \lambda^* - \Delta\lambda \leq \lambda \leq \lambda^* + \Delta\lambda, \quad \beta \geq 0. \end{aligned} \quad (3)$$

1) *Adapt Wavelength  $\lambda$* : Similar to the initial training, we allow local adjustment of wavelength with  $\pm 20\text{nm}$  range to adapt the diffraction operation to new tasks.

2) *Adapt Metasurface Spacing  $z$* : Similar to the initial training stage, we can either learn a shared spacing  $z \in \mathbb{R}$  or a dedicated layer-wise spacing vector  $\mathbf{z} \in \mathbb{R}^L$  with the standard gradient descent method.

3) *Learn Metasurface Orientation  $\mathcal{O}$  with Gumbel-Softmax Method*: Selecting the orientation of metasurfaces is a highly discrete problem. We model the orientations of each metasurface as a 9-category parametric distribution. For the  $(p, l)$ -th metasurface, its probability of having orientation  $\mathcal{O}_{p,l} \in \mathbf{O} = \{N, W, S, E, FN, FW, FS, FE, BP\}$  is

$$P_{\mathcal{O}_{p,l}}(\mathcal{O}_{p,l} = \mathbf{O}[i]) = e^{\theta_{p,l}^i} / \sum_{i=1}^9 e^{\theta_{p,l}^i}. \quad (4)$$

We employ Gumbel-softmax to learn the distribution parameter  $\theta$  with a gradient-based method in Fig. 3. The phase mask  $\Phi$  is the superposition of all 9 states weighted by a mask  $m$ , i.e.,  $\Phi_{p,l} = \sum_{i=1}^9 m_{p,l}^i \Phi_{p,l}^i$ , where the mask is sampled from the following distribution,

$$m_{p,l}^i = \text{GumbelSoftmax}(\theta_{p,l}^i | \mathcal{O}_{p,l}) = \frac{e^{(\theta_{p,l}^i + g_{p,l}^i)/\tau}}{\sum_i e^{(\theta_{p,l}^i + g_{p,l}^i)/\tau}}, \quad (5)$$

where  $g$  and  $\tau$  are the Gumbel noise and temperature. The final orientation  $\mathcal{O}_{p,l}$  can be selected by  $\mathcal{O}_{p,l} = \mathbf{O}[\text{argmax}_i \theta_{p,l}^i]$ .

4) *Learn Metasurface Placement Order  $\mathcal{P}$  with Differentiable Permutation Learning*: We introduce an automatic learning method to permute all  $PL$  metasurfaces in a differentiable way. Without loss of generality, we use the learning of a permutation across  $P$  metasurfaces as an example. Exploring the factorial permutation space is very challenging. Inspired by a differentiable circuit routing learning algorithm [23], as illustrated in Fig. 5, we use a  $P \times P$  permutation matrix  $\mathcal{P}$  to represent the mapping from metasurface indices to slot indices. A permutation matrix is a Boolean matrix  $\mathcal{P} \in \{0, 1\}^{P \times P}$  where each row and column only has one 1.  $P_{ij} = 1$  means the  $i$ -th

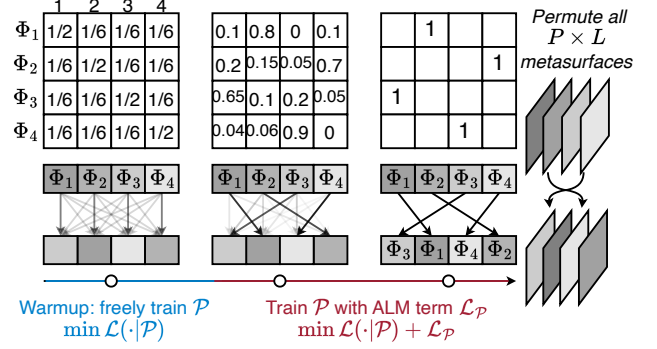


Fig. 5: Illustration of metasurface placement order permutation. Two-stage training is adopted with warmup training and ALM-based permutation learning.

metasurface is placed at slot  $j$ . We convert the permutation constraint with its equivalent continuous format, i.e., the row-wise and column-wise  $\ell_1$ -norm equals the squared  $\ell_2$ -norm. We solve the constrained optimization problem with the augmented Lagrangian method (ALM) to relax the constraint as an objective term,

$$\mathcal{L}_{\mathcal{P}} = \sum_{i=1}^P \mu_{i,r}^T (\Delta \tilde{\mathcal{P}}^{i,:} + \frac{\rho}{2} (\Delta \tilde{\mathcal{P}}^{i,:})^2) + \mu_{i,c}^T (\Delta \tilde{\mathcal{P}}^{:,j} + \frac{\rho}{2} (\Delta \tilde{\mathcal{P}}^{:,j})^2), \quad (6)$$

where  $\mu^r \in \mathbb{R}^P$  and  $\mu^c \in \mathbb{R}^P$  are row-wise and column-wise Lagrangian multipliers, and  $\rho$  is the quadratic penalty weight.  $\Delta \tilde{\mathcal{P}}^{i,:} = \|\tilde{\mathcal{P}}^{i,:}\|_1 - \|\tilde{\mathcal{P}}^{i,:}\|_2^2$ . At every step when we update  $\mathcal{P}$ , we also perform the dual update on the multiplier  $\mu \leftarrow \mu + \rho(\Delta \tilde{\mathcal{P}}^{i,:} + \frac{1}{2}(\Delta \tilde{\mathcal{P}}^{i,:})^2)$  to gradually increase the penalty to push it to a feasible permutation.

5) *Train Task-specific Head Layers*: After the backbone, task-specific head layers, such as the final linear classifier or convolutional regression layers, are re-initialized and trained on the new task.

## V. EVALUATION RESULTS

### A. Settings

1) *Model and Dataset*: As a case study, we assume our CHORD system has up to 4 paths and up to 4 cascaded metasurface layers. The DONN model backbone based on this hardware contains two Hybrid DONNBlocks, with depthwise DiffLayers mapped to the same CHORD hardware. All structural and mechanical parameters that cannot be switched in runtime, i.e.,  $\lambda$ ,  $z$ ,  $s$ ,  $\Phi$ ,  $\mathcal{O}$ ,  $\mathcal{P}$ , are shared across all DiffLayers. Runtime reconfigurable parameters, i.e.,  $\alpha_{pre}$ ,  $\alpha_{post}$ , and  $\beta$ , are independently learned for each DiffLayer. Each metasurface has  $32 \times 32$  meta-atoms. The initial values for wavelength  $\lambda = 532 \text{ nm}$ , meta-atom size  $s = 400 \text{ nm}$ , and metasurface spacing  $z = 8.42 \mu\text{m}$  are typical values used in the literature [17]. We evaluate on three benchmarks:

- Pre-fab training on Fashion-MNIST [24] 10-class recognition and adapt the model to QuickDraw [25] 10-class image recognition. A classification head is used with AdaptiveAvgPool5-FC10.
- Pre-fab training on CIFAR-10 [26] and adapt the model to CIFAR-100 [26]. The classification head is Adaptive AvgPool5-FC10(100).
- Pre-fab training on Darcy flow [27] 1-channel PDE solving and adapt to Navier-Stokes [28] 10-channel PDE solving. A regression head is used with C64K3-BN-GELU-C1(10)K1.

2) *Training Settings*: General Setting: We train all DONNs for 100 epochs using the AdamW [29] optimizer with a learning rate (lr) of  $1\text{E-}2$ , a cosine decay scheduler, a weight decay rate of  $1\text{E-}2$ , and data augmentation techniques such as random cropping and flipping for both

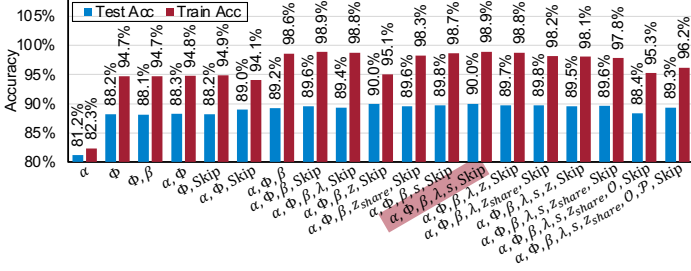


Fig. 6: Expressivity of different combinations of learnable system parameters. Training and test accuracy are shown on 2-layer DONN on FMNIST.

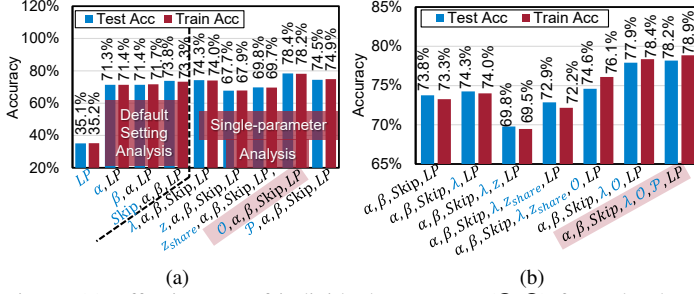


Fig. 7: (a) Effectiveness of individual parameter (2-6) for task adaptation, LP: Linear Probing. (b) Effectiveness of progressively enabling parameters (2-6) in adaptation.

pre-fabrication training tasks and adaptation tasks. Weight decay is only applied to  $\Phi$ , which is reparameterized in the range of  $[-\pi, \pi]$ .

**Schedule  $\tau$  in Orientation and  $\rho$  in Permutation:** For orientation learning, to prioritize early-stage exploration and later-stage stable convergence, we apply exponential decay on the Gumbel-softmax temperature  $\tau = \tau_{init} \cdot \gamma_{\tau}^{(epoch-1)}$ , where  $\tau_{init}$  is 5 and  $\gamma_{\tau}$  is 0.956. To encourage early-stage free permutation optimization and gradually force the constraints, ALM quadratic penalty term  $\rho$  is scheduled as  $\rho = \rho_{init} \cdot \gamma_{\rho}^{(epoch-warm\ up\ epoch)}$ , with a warm-up period of 5 epochs,  $\rho_{init}$  set to  $1E-5$ , and  $\gamma_{\rho}$  set to 1.2.

**Parameter Quantization:**  $\lambda$  can be adjusted within  $\pm 20$  nm around initial value, with 1 nm precision. To ensure that the meta-atom can generate a  $-\pi$  to  $\pi$  phase response for a given  $\lambda$ , the pixel size's lower bound is calculated following Sec. IV-A3 equation:  $s > \frac{\lambda}{\lambda_0} \cdot s_{2\pi, \lambda_0} + \Delta_s$ , where  $\lambda_0 = 532$  nm,  $s_{2\pi, \lambda_0} = 300$  nm, and  $\Delta_s = 20$  nm. The precision for  $z$  is set to 10 nm, while  $\Phi$  is reparameterized  $\in [-\pi, \pi]$  with a precision of  $1^\circ$ . We also employ learned step-size quantization-aware training [30]. A  $b_w=8$ -bit quantization is applied to weights and activations in layers utilizing PTCs, such as  $\alpha$  and the final head.

**Evaluation Metrics:** Accuracy is used as the primary metric for the classification tasks. The Mean Squared Error (MSE) was utilized as the evaluation metric for PDE-solving tasks. Both training and test performance are shown to indicate each setting and parameter's contribution to the DONN's expressivity, trainability, and generalizability.

## B. Ablation Study on CHORD's Learnable Parameters

1) **Parameter Expressivity and Learnability in Pre-fab Training:** In initial training, we conduct a thorough ablation study on parameters in Fig. 6. A single-path DONN is assumed ( $P = 1, L = 4$ ). We train ( $\alpha, \Phi, \beta, \lambda, s$ ) with the optical skip path enabled as the optimal default pre-fab training settings.

2) **Individual Effectiveness of Params. 2-7 in Post-fab Task Adaptation:** We explore the impact of each parameter on task adaptation

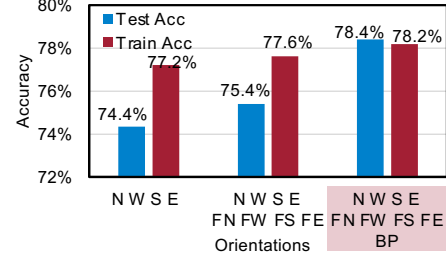


Fig. 8: Nine orientations show the best expressivity and accuracy on QuickDraw adaptation tasks.

TABLE II: Evaluate CHORD ( $P=4, L=4$ ) on pre-fab initial tasks and post-fab adapted tasks. A 2-layer digital CNN ( $C32K3$ )<sub>2</sub> is shown as a reference. In each group, DONN is trained on the first task and adapted to the second task. We use initial training results of the second task as a reference. For classification, test Acc. is shown. For Darcy/NavierStokes, MSE is shown.

| Tasks        | Digital 2-Layer CNN<br>C32K3-C32K3 | Pre-fab<br>Initial Training<br>$P=4, L=4, \alpha, \Phi,$<br>$\beta, Skip, s, \lambda$ | Post-fab<br>Task Adaptation<br>$P=4, L=4, \alpha, \beta, Skip,$<br>$\lambda, O, P$ |
|--------------|------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| FMNIST       | 0.9122                             | 0.9014                                                                                | -                                                                                  |
| Quickdraw    | 0.8978                             | 0.8646                                                                                | 0.7992 (adapt from FMNIST)                                                         |
| CIFAR10      | 0.7633                             | 0.6182                                                                                | -                                                                                  |
| CIFAR100     | 0.4347                             | 0.3815                                                                                | 0.2513 (adapt from CIFAR10)                                                        |
| Darcy        | 0.0466                             | 0.055                                                                                 | -                                                                                  |
| NavierStokes | 0.1085                             | 0.1007                                                                                | 0.1753 (adapt from Darcy)                                                          |

in Fig. 7(a). Simply training the head, called linear probing (LP), with the backbone frozen, is insufficient. As we enable the optimization of electrically tunable parameters ( $\beta$  and  $\alpha$ ), along with the optical skip path and linear probing, we observe a significantly enhanced accuracy.

**Mechanically tunable parameters ( $\lambda, z, O, P$ )** further improve task performance, with phase mask orientation  $O$  showing the largest impact due to its additional  $9^4 \times$  larger solution space. Permutation  $P$  also contributes positively to task adaptation, though not as impactful as orientation due to the  $4!$  solution space with only 4 metasurfaces (single path),  $273 \times$  smaller than  $O$ . The effectiveness of  $P$  is expected to factorially grow as the number of paths increases.

The diffractive distance  $z$  and  $z_{share}$  contribute less to performance, as they are too sensitive to be trained alongside other parameters, leading to an increase in training difficulty and thus impacting the overall optimality, which is not trained during adaptation.

3) **Combinational Effectiveness of Params. 2-7 in Post-fab Task Adaptation:** Building on the individual parameter analysis, we progressively enable these parameters to evaluate their combined effectiveness in task adaptation, as shown in Fig. 7(b). The results are consistent with those in Sec. V-B2, confirming that the diffractive distance  $z$  is highly sensitive and tends to reach suboptimal states when trained alongside other parameters. However, fixing  $z$  to its initial learned value significantly improves adaptation accuracy as additional parameters,  $\lambda, O$ , and  $P$ , are progressively introduced.

4) **Orientation Effectiveness:** Figure 8 provides strong evidence for the effectiveness of the 9-option orientations. Compared to 4-direction rotation [13], [14], the inclusion of 4 flipped states and the identity state improves test accuracy by +4%, demonstrating their significant contribution to CHORD's task adaptability.

## C. Multi-Path, Multi-Layer Exploration

We investigate the impact of multiple parallel paths in CHORD, exploring a more generic design space to enable greater functionality.

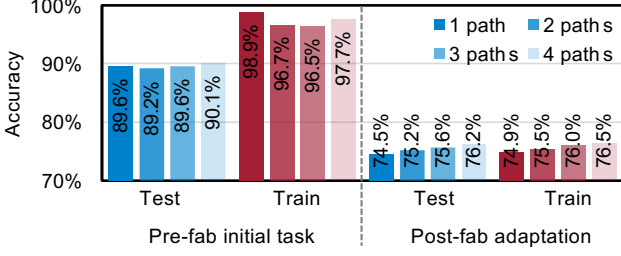


Fig. 9: Adapt four DONNs with  $L=4$  and different paths  $P$  from pre-fab FMNIST task to post-fab Quickdraw task. Only permutation  $\mathcal{P}$  is trained during adaptation to show the learnability gain by enlarging the solution space from  $4!$  to  $16!$ .

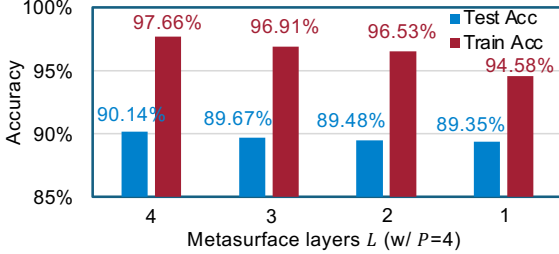


Fig. 10: Different layers  $L$  with  $P=4$  on pre-fab FMNIST.

1) *Number of Paths  $P$  in Pre-Fab Initial Training:* In Fig. 9, we fix the layer count to  $L=4$  and increase the number of diffractive paths  $P$ . Test accuracy on the initial task improves by approximately 0.5% with an increasing number of paths. However, fluctuations in training accuracy suggest that adding too many parallel phase masks, similar to large-kernel convolutions, introduces learning challenges.

2) *Number of Paths  $P$  in Post-Fab Task Adaptation with Permutation:* In Fig. 9(Right), we show task adaptation only with permutation learning enabled. By increasing the path from 1 to 4, the space of permutation increases from  $4!$  to  $16!$ , leading to a notable learnability boost and thus +1.7% improvement on test and training accuracy.

3) *Number of Layers  $L$  in Pre-Fab Initial Training:* We evaluate the impact of layer depth  $L$  on learnability by conducting pre-fab training with 4 parallel paths and varying the number of layers  $L$ . Figure 10 shows a significantly enhanced expressivity with more layers cascaded, demonstrating the clear benefit of deeper CHORD.

#### D. Main Results

We conduct three pre-fab training tasks on Fashion-MNIST, CIFAR-10, and Darcy, and transfer the CHORD system to new tasks, i.e., QuickDraw, CIFAR-100, and Navier-Stokes, respectively. Table II demonstrates that CHORD exhibits competitive expressivity compared to the digital counterpart in FMNIST and QuickDraw. However, we observe accuracy gaps on CIFAR-10 and CIFAR-100 due to the increased difficulty of those tasks. Notably, CHORD **excels in much harder PDE prediction tasks such as Darcy flow and Navier-Stokes. These tasks benefit from a global receptive field, aligning perfectly with the global spatial processing capabilities of diffractive ONNs.**

For task adaptation, when all system variables ①-⑥ are enabled to learn, we can largely resume the accuracy even when the major weights  $\Phi$  are frozen. To understand the individual contribution of each variable to task adaptability, we show a detailed discussion below.

#### E. Discussion

1) *Parameter Contribution to Adaptability:* Figure 11 shows the insights of parameter contribution to adaptability.

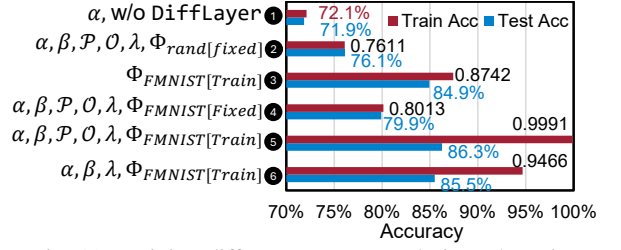


Fig. 11: Training different parameters during adaptation.

TABLE III: Device info in energy and latency calculation [12].

| Device     | 8-bit ADC | 8-bit DAC | TIA | MZM | PS   | PD |
|------------|-----------|-----------|-----|-----|------|----|
| Power (mW) | 14.8      | 50        | 3   | 50  | 0.15 | 0  |

Compare (①, ④): Besides  $\alpha_{pre}$ ,  $\alpha_{post}$  on integrated photonics, DiffLayers are also critical in DONN learning.

Compare (②, ④): The trained  $\Phi$  from initial tasks carries transferable knowledge that helps new task learning.

Compare (④, ⑥): In the future, with advancements in *reconfigurable metasurfaces*, enabling  $\Phi$  to be trainable on new tasks post-fabrication could significantly enhance performance, unlocking even greater potential for dynamic task adaptability and learnability.

Compare (③, ⑥): We want to highlight that even if  $\Phi$  becomes trainable in the future, our introduced  $\alpha, \beta, \mathcal{P}, \mathcal{O}, \lambda$  parameters can still enhance the network's expressivity and learnability.

Compare (⑥, ⑥): It validates that our introduced permutation  $\mathcal{P}$  and orientation  $\mathcal{O}$  effectively expand the solution space and are crucial for enhancing the model's adaptability.

2) *Training Efficiency of Our Depth-wise DiffLayer:* Our depth-wise DiffLayer achieves significant efficiency improvements over prior DONNs with 64-channel Conv-like diffractive layers, reducing memory by 29% (from 9807 MB to 6997 MB) and accelerating training by  $4.92\times$  per iteration (from 329.59 ms to 66.99 ms).

3) *Energy and Speed Advantages of CHORD vs. Digital CNNs:* Via behavior-level architecture simulation, on average, CHORD delivers  $74\times$  faster single-batch inference speed (1.48 ms to 0.02 ms) and consumes  $194\times$  less energy (134.9 mJ to 0.696 mJ) compared to a digital CNN on A6000 GPU. For the DONN part, we assume a photodetection (PD) speed of 10 GHz [31]. Given that  $P$  channels are processed in parallel, this results in an  $8E-7$  ms latency with  $32/P=8$  ( $P=4$ ) cycles and 0.1 ns per cycle. Thus, the latency of the diffractive layer is negligible compared to the overall time.

For the PTC part, we employ a single-core  $16\times 16$  PTC design [12], with device power listed in Table III. The energy and latency are simulated using an open-source photonic accelerator simulator [20].

## VI. CONCLUSION

We present the first in-depth analysis of multi-dimensional learnability in DONNs, introducing a physically composable hybrid optical/photonic system, CHORD. Our results demonstrate that CHORD with multi-dimensional learnable system variables provides digital-comparable accuracy on various task adaptations, especially challenging PDE-solving tasks, with  $74\times$  faster inference speed and  $194\times$  lower energy. Compared to prior DONNs, our hybrid CHORD shows exponentially larger functional space and extreme multi-functionality with  $5\times$  faster training speed. CHORD shows the potential to enable versatile and high-efficiency optical AI systems with repurposed hardware, paving the way for practical deployment in dynamic AI applications.

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